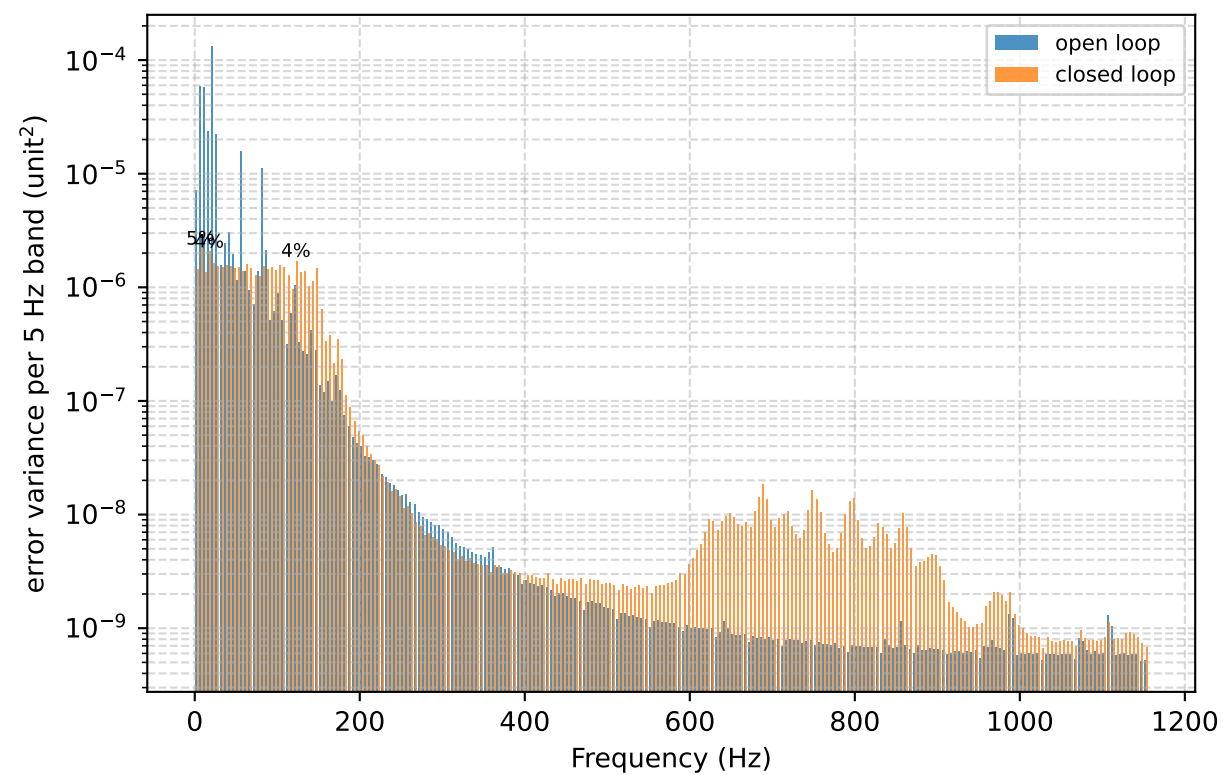
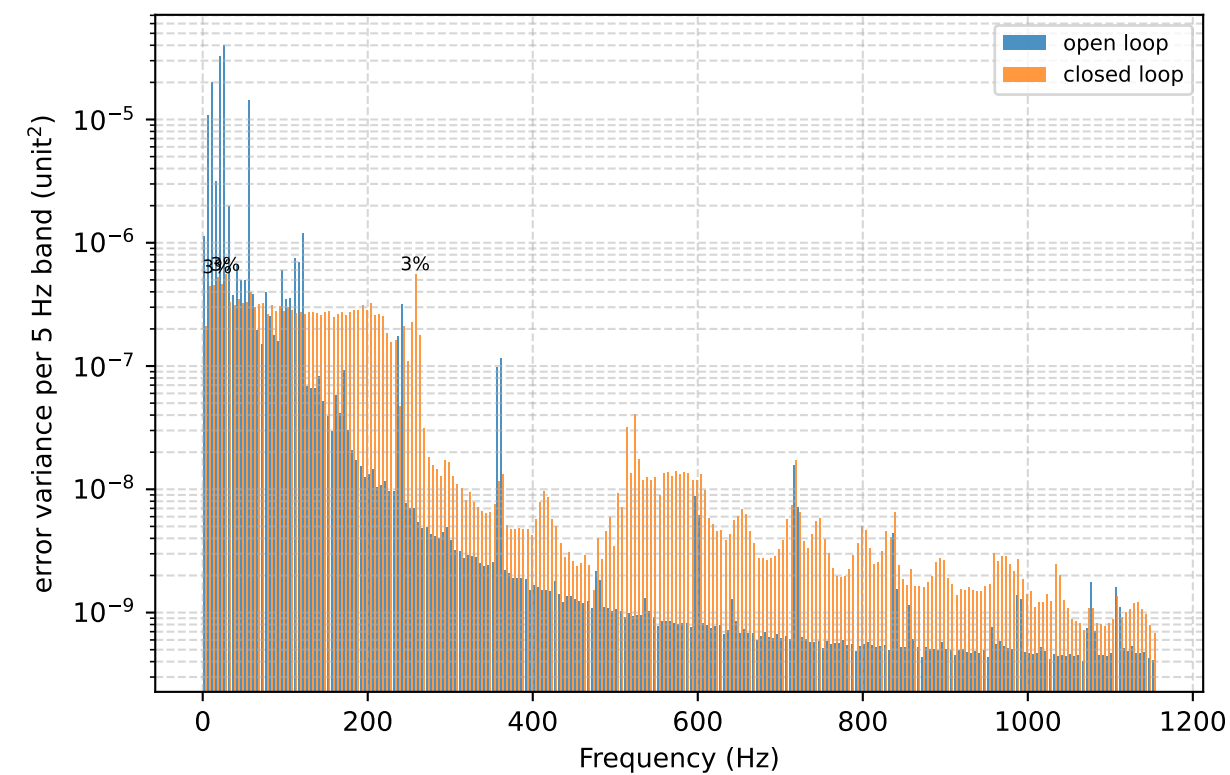
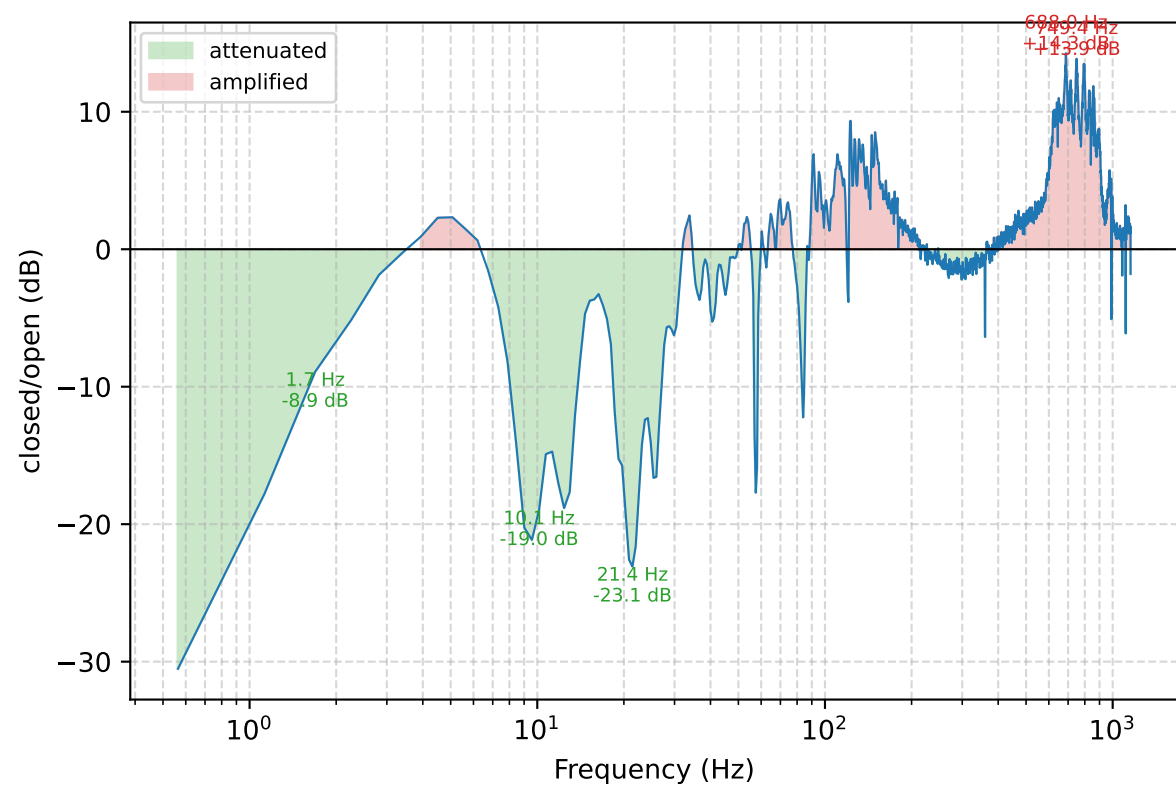
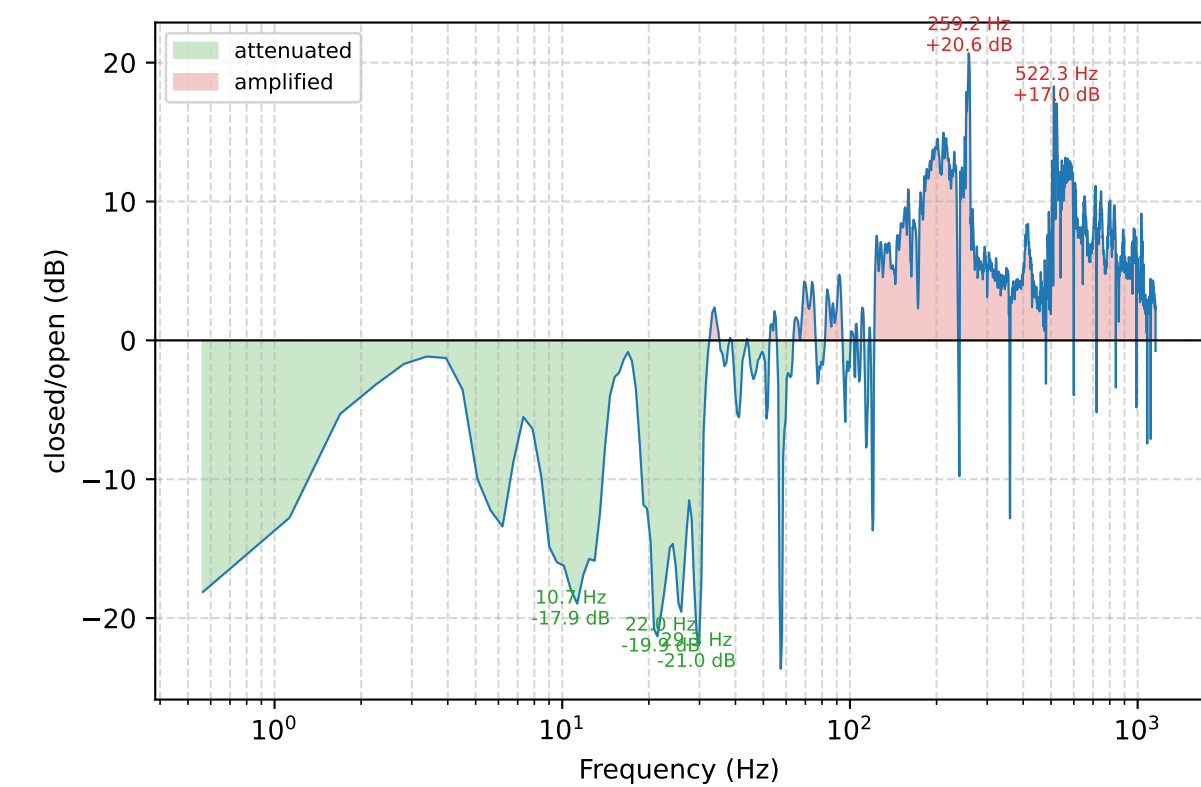
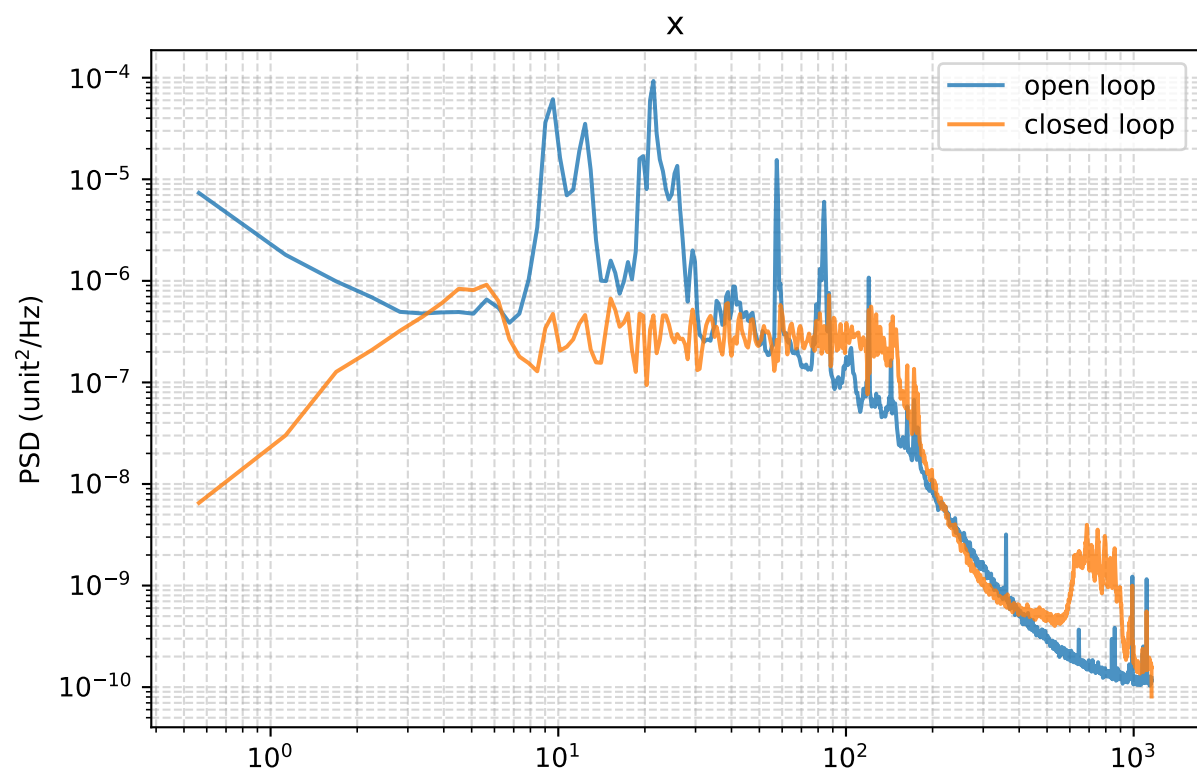
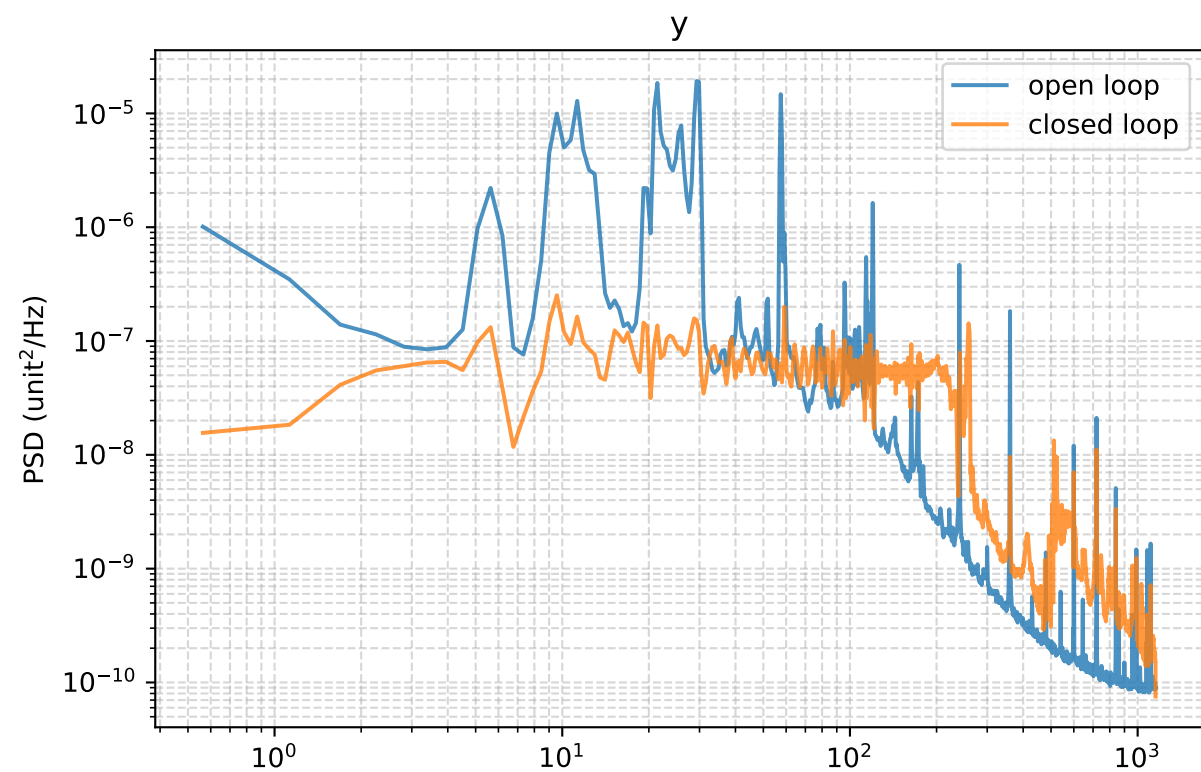


Closed-loop attenuation



Run configuration and per-band RMS

Started: 2026-07-15 13:30:40
pixel_scale: 31.5 px per output unit
y: mean -0.044 -> +0.001 px; RMS about mean 0.399 -> 0.127 px (open -> closed)
x: mean -4.191 -> +0.000 px; RMS about mean 0.699 -> 0.218 px (open -> closed)

Config used:
FULL-BAND FSP (Smith reconstruction + identified Wiener/Kalman predictor), replaces pid+kalman
latest Bode delay=1.81 ms=4.18 frames; discrete delay=4, fs=2310, broad_order=512, broad_mu=.03
The broad observer supplies the command; AR2 modes remain available for diagnostics/adaptation and are not summed.
attenuation10 open-PSD prediction bound: delay 4.0 y=0.0617 x=0.1026 px; delay 4.18 y=0.0644 x=0.1071 px
accept only after a >=500 s closed hardware run with no command clipping or HF instability
refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.1092	0.0255	0.2562	0.0602
10-20	0.1525	0.0312	0.2837	0.0582
20-30	0.2675	0.0318	0.3939	0.0561
30-40	0.0482	0.0252	0.0632	0.0551
40-50	0.0338	0.0259	0.0705	0.0547
50-60	0.1217	0.0269	0.1305	0.0538
60-70	0.0240	0.0248	0.0480	0.0552
70-80	0.0234	0.0241	0.0458	0.0501
80-90	0.0207	0.0242	0.1147	0.0544
90-100	0.0275	0.0240	0.0335	0.0540
100-110	0.0264	0.0241	0.0374	0.0553
110-120	0.0380	0.0232	0.0301	0.0471
120-130	0.0354	0.0231	0.0368	0.0552
130-140	0.0115	0.0232	0.0230	0.0490
140-150	0.0116	0.0230	0.0263	0.0511
150-160	0.0083	0.0229	0.0160	0.0312
160-170	0.0099	0.0231	0.0157	0.0243
170-180	0.0111	0.0231	0.0171	0.0241
180-190	0.0061	0.0237	0.0116	0.0141
190-200	0.0052	0.0242	0.0095	0.0110
200-210	0.0053	0.0240	0.0085	0.0095
210-220	0.0046	0.0226	0.0079	0.0080
220-230	0.0046	0.0184	0.0071	0.0070
230-240	0.0135	0.0144	0.0063	0.0059
240-250	0.0179	0.0178	0.0058	0.0056
250-260	0.0037	0.0279	0.0054	0.0048
260-270	0.0032	0.0145	0.0050	0.0043
270-280	0.0030	0.0058	0.0045	0.0038
280-290	0.0028	0.0052	0.0042	0.0036
290-300	0.0031	0.0058	0.0040	0.0034
total 0-300	0.3633	0.1246	0.5934	0.2160